

Student Grade System – Python Code

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Explorer
  GenAI Program
  venv
  > Include
  > Lib
  > Scripts
  .gitignore
  pyvenv.cfg
  grade_system.py

Welcome X grade_system.py X
  grade_system.py > main
1  # Student Grade System
2  # CAIE Course Program - Social Eagle
3
4  def calculate_grade(mark):
5      """Return the letter grade for a valid mark."""
6      if mark >= 90:
7          return "A"
8      elif mark >= 80:
9          return "B"
10     elif mark >= 70:
11         return "C"
12     elif mark >= 60:
13         return "D"
14     else:
15         return "E"
16
17
18  def main():
19      try:
20          #Asks the user to enter a mark. Converts the input from text to a decimal number using float().
21          mark = float(input("Enter your mark (0-100): "))
22
23          # Validate the mark range - Checks that the mark is between 0 and 100.
24          if mark < 0 or mark > 100:
25              print("Invalid mark. Please enter a number between 0 and 100.")
26              return
27          #Calls calculate_grade(mark).
28          grade = calculate_grade(mark)
29
30          # Display the result - Prints the mark and resulting grade.
31          print(f"Mark: {mark:g} -> Grade: {grade}")
32      #Handles invalid non-numeric input using except ValueError.
33      except ValueError:
34          print("Invalid input. Please enter a valid number.")
35
36  #ensures main() runs only when this file is executed directly, not when it is imported into another Python file.
37  if __name__ == "__main__":
38      main()
```

Output Results -Screenshot :-

```
File Edit Selection View Go Run Terminal Help
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Problems Output Debug Console Terminal Ports
(venv) C:\Users\Adin\GenAI Program>python grade_system.py
Enter your mark (0-100): 90
Mark: 90 -> Grade: A

(venv) C:\Users\Adin\GenAI Program>python grade_system.py
Enter your mark (0-100): 50
Mark: 50 -> Grade: E

(venv) C:\Users\Adin\GenAI Program>python grade_system.py
Enter your mark (0-100): 120
Invalid mark. Please enter a number between 0 and 100.

(venv) C:\Users\Adin\GenAI Program>python grade_system.py
Enter your mark (0-100): hello
Invalid input. Please enter a valid number.

(venv) C:\Users\Adin\GenAI Program>
```

Github Respository Screenshot:-

[prasanna1280/GENAI_PROGRAM](https://github.com/prasanna1280/GENAI_PROGRAM): This Repository contains of GENAI Architect program (Social Eagle AI) of all weeks and days assignments code including documents

The screenshot displays the GitHub repository page for **prasanna1280/GENAI_PROGRAM**. The repository is public and contains 1 branch and 0 tags. The main branch is **main**. The repository description states: "This Repository contains of GENAI Architect program (Social Eagle AI) of all weeks and days assignments code including documents".

The repository contains the following files:

File Name	Commit Message	Commit Time
Python_VScode_GitHub_installation.pdf	Week1_Day1_Assignment	3 minutes ago
README.md	Initial commit	6 minutes ago
grade_system.py	Week1 - Day2 Assignment	now

The README section is titled **GENAI_PROGRAM** and contains the text: "This Repository contains of GENAI Architect program (Social Eagle AI) of all weeks and days assignments code including documents".

The right sidebar shows the repository's activity, including 0 stars, 0 watching, and 0 forks. It also lists the releases and packages sections, both of which are currently empty.

Hand Notes on Python concepts -

